



The here and now of climate change: Climatic trends throughout Australian cotton regions and implications for the growing season

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ABSTRACT

Context: Much research has focused on assessing future climatic conditions on cotton production. However, understanding climatic changes that have already occurred throughout Australian cotton regions may provide insights on how existing and new regions are adapting to change.

Objectives: This study aims to understand climatic changes that have occurred throughout Australian cotton regions, and the implications that these changes may have had for cotton growth and development.

Methods: We utilised historical weather data to undertake an analysis of regional climate using changes in: heat unit accumulation for seasonal assessment of temperature conditions; frequency of extreme weather conditions affecting cotton growth; seasonal potential evapotranspiration (ET_0); and the timing of key cotton developmental stages. Nine Australian cotton regions over three time periods from 1961 to 2020 were assessed. We also used the cotton simulation model OZCOT to predict yield, crop evapotranspiration (ET_C), and water use efficiency (WUE, lint yield kg/mm ET_C) for a crop grown at each location.

Results: Our data shows some reductions in early season cold shocks ($\leq 11^\circ\text{C min.}$), and increases in minimum temperature, seasonal heat accumulation, ET_0 , and the frequency of extremely high temperatures ($> 40^\circ\text{C}$). Overall, time to first flower (FF) has decreased across several locations which may be having implications for less leaf area prior to flowering, but it may be contributing to an opportunity for longer reproductive periods at Emerald and Moree.

Conclusions: Climatic changes including decreased frequency of early season cold shocks, increased minimum temperatures and frequency of high temperatures and evapotranspiration, and an increase in season long heat accumulation have occurred across Australian cotton regions from 1961 to 2020, affecting growth and development of cotton. Overall, warmer seasonal temperatures have reduced the time to first flower, increased the time to last effective flower (LEF), and resulted in some changes to last harvestable boll (LHB).

Implications: Climatic changes have resulted in an opportunity for longer reproductive periods and the potential for increased yields. Recognising that regions have experienced specific changes may help to guide current and future agronomic practice in the different regions.

1. Introduction

In Australia, cotton is predominately grown throughout central and southern Queensland and northern New South Wales (NSW), with recent expansion of the industry into southern NSW. In these production systems, cotton is managed as an annual crop, planted in September/October and harvested in March/April. The climatic zones across major Australian cotton regions are defined as sub-tropical to warmer temperate zones (McMahon and Low, 1972). Yields are determined by a

combination of environmental, management and genetic factors, which is important to bear in mind when considering the past and projected effects of climate change on agricultural systems. Yield gains of cotton are attributed to plant breeding and agronomic management strategies (Liu et al., 2013). In a 30-year evaluation of cotton breeding trials (from 1980 to 2009), genetics (i.e., cultivar, 48 %), management (28 %) and cultivar $\times$ management interaction (24 %) contributed to yield gains. Liu et al., (2013) acknowledged that some of the increases in the management component would be attributed to increases in atmospheric

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CO₂ concentrations over time. In this period CO₂ had increased in concentration by 49 ppm (Tans and Keeling, 2023).

Australia's climate has warmed on average $1.47^{\circ}\text{C} \pm 0.24$ since 1910, leading to an increase in the frequency of extreme heat events (CSIRO and Bureau of Meteorology, 2022), and an increased severity of drought conditions during periods of below-average rainfall. Warmer temperatures are observed across Australia in all seasons with both day and night-time temperatures increasing, and with the shift to a warmer climate accompanied by more extreme daily heat events. There have been few studies specifically exploring how the climate has already changed across Australian cotton regions and how these changes relate to cotton growth, development, and production practice. Understanding these may help to define the adaptation strategies and develop an agenda for sustainable and profitable Australian cotton production in a changing climate. Where there are similarities across regions, there also may be an opportunity to apply knowledge to better manage crops and optimise production for changing climatic scenarios.

Temperature plays many important roles in the growth and development of cotton. Optimal plant temperature for cotton physiology and yield is $28 \pm 2^{\circ}\text{C}$ (Conaty et al., 2012), however air temperatures vary greatly both across the different cotton regions and throughout the season. Minimum daily temperatures $< 11^{\circ}\text{C}$, referred to as “cold shocks”, increases the time to emergence and reduces seedling vigour, which may lead to poor establishment, poor early growth, and increased risk of seedling diseases, thus delaying the development of cotton seedlings (Bange and Milroy, 2004b; Constable, 1976). High daytime temperatures ($> 35^{\circ}\text{C}$) impairs flower survival, reduces fruit retention and consequently limits yield (Reddy et al., 1992; Reddy et al., 1991a). Cotton development can be predicted using seasonal temperature converted to day degrees (DD). DD is the accumulation of heat units calculated from the daily minimum and maximum temperatures. Management recommendations are based on the timing of crop phenology, commonly estimated by growing day degrees (Bange et al., 2022). At the end of the cotton season, cooler temperatures will affect the timing of crop maturity and influence the effectiveness of chemical harvest aids, directly affecting yield and quality. Late in the season cold temperature ($< 2^{\circ}\text{C}$) can signal the end of the growing season.

Luo et al. (2014) conducted a modelling study to understand projected climate change throughout Australian cotton production regions, and to quantify its impacts on cotton crop phenology and the occurrence of heat stress and cold shocks in the cotton growing season, using climate models (CCAM). Results showed that there will be less impact of cold temperatures and a longer growing season that may be beneficial for cotton production. However, there will be more incidences of heat stress impacting growth, and more rapid crop development and maturity that may limit cotton to capitalise on the opportunities associated with increases in growing season length, which may require adjustment in management strategy. Welsh et al. (2022) reviewed historical yield and crop production methods, accounting for seasonal climatic conditions. Their study explored the effects of climate drivers of ENSO (El Niño Southern Oscillation), IOD (Indian Ocean Dipole) and SAM (Southern Annular Mode), and their correlation to cotton lint yield in dryland and irrigated systems across cotton regions. However, the influence that climatic changes have had on cotton development across regions was not specifically assessed.

Climate change will have both positive and negative effects on cotton production (Bange et al., 2016; Broughton et al., 2017b). Increased CO₂ may increase yield in well-watered crops and higher temperatures will extend the length of the growing season. However, warmer temperatures also accelerate the rate of crop development and could potentially shorten the time to maturity, which may then impact crop management decisions. Higher temperatures also have the potential to cause significant fruit loss, reduce water use efficiencies, lower yields, and alter fibre quality. This study aims to understand climatic changes that have occurred throughout Australian cotton regions and the implications of a changing climate on cotton development. We tested the hypotheses that

air temperature has increased across nine key cotton growing regions, leading to an acceleration of crop development to first flower (FF), while a delay in the last effective flower (LEF) and last harvestable boll (LHB), and consequently increasing reproductive period underpinning cotton yield.

2. Materials and method

2.1. Assessing trends in climate change across cotton regions

Nine representative locations spanning cotton production areas in Queensland (Emerald (most northern location -23° Lat 148° Long), Dalby, and Goondiwindi) and New South Wales (Moree, Narrabri, Gunnedah, Narromine, and Griffith (most southern location -34° Lat 146° Long) were assessed. For a map of Australian cotton regions, please refer to: <https://cottonaustralia.com.au/where-is-cotton-grown>. Seasonal data from 1959 to 2022 at each location was retrieved from the SILO (<https://www.longpaddock.qld.gov.au>) gridded dataset based on a longitude and latitude for the center of the nearest town. When necessary, this used interpolated data from the surrounding weather stations to reflect the regional weather more accurately. Climate data for each cotton season was assessed between 1st September and 30th April. Normal planting time across Australian cotton regions for OZCOT simulations are detailed in Luo et al. (2016). For each planting year linear regression lines for three time periods (a) 1961–2020; (b) 1961–1990; and (c) 1991–2020 were used to assess the trend in key growing season climate variables for each period. These periods and lengths were selected based on the climate normal approach of Arguez and Vose (2011), selecting two 30 year periods covering from early in cotton production in Australia to the present day.

Trends during the above periods were assessed for the following variables were assessed for their change over these periods:

- Temperature extremes – Average minimum air temperature during the season, the frequency of cold shock events (number of days between emergence and first flower where the minimum air temperature is less than 11°C), and frequency very hot days, where the maximum air temperatures exceed 40°C and 45°C .
- Seasonal day degree accumulation - Air temperature was also used to calculate the total day degrees from the beginning to end of season (1st September to 30th April) using a base of 12°C (Constable and Shaw, 1988) (DD12) and a base of 15.6°C and an optimum threshold of 32°C (Bange et al., 2022) (DD1532). The method of Constable and Shaw (1988) was until recently the common approach used by the Australian industry and is the method that has been used in future climate change studies (Luo et al., 2014). It was acknowledged in this study that consideration of an optimum temperature threshold should be considered given a warming environment. The method of Bange et al. (2022) has been recently adopted by industry as the standard for assessing and predicting crop development, however, it is important to note that the DD1532 approach does not include the negative effect of cold shock on cotton.
- Total potential evapotranspiration ET_O and rainfall (mm) were assessed for the same season length mentioned above. ET_O throughout the cotton season reflects atmospheric evaporative demand, which influences crop water use.
- Predictions of cotton development phases - Cotton phases were calculated using degree days with a base 12°C . First flower (FF) was predicted at 777DD after sowing. The last harvestable boll (LHB) was calculated based on the first autumn occurrence of frost (air temperature minimum $< 2^{\circ}\text{C}$). The last effective flower (LEF) was back calculated as 750DD (a boll period) before LHB. The time of first flower and LEF represents the length of the reproductive period; a longer phase usually equates to more yield (Bange and Milroy, 2004a). The timing of LHB affects the timing of harvest; a later date can mean that bolls have more time to fill and is related to the timing

of LEF. LEF is considered “cut-out” in cotton production. Cut-out occurs when a plant stops producing new fruit, as the high demand for assimilates by developing fruit leaves none available for the initiation of new fruit (Bange and Milroy, 2000).

Seasonal accumulation of DD reflects the length of the growing season, frequency of cold and heat events represents extreme effects on growing conditions.

2.2. Yield Modelling using the OZCOT crop simulation model

The cotton production model OZCOT version circa 2018 (Hearn, 1994) was used to predict yield, evapotranspiration (ET), flowering dates and water use efficiency (WUE) for a crop grown at each location. The central component of the OZCOT model is the fruit production and survival subroutine (Hearn and Da Roza, 1985). The rates of fruit production, fruit shedding and growth of the organs are governed by carbon supply. Carbon supply for a given day is estimated from intercepted light and crop level photosynthetic rate with respiration deducted. Light interception is estimated, and leaf area generated using an empirical correlation between fruiting site production and leaf area. The leaf expansion rate, photosynthesis and fruiting are modulated by the water supply and nitrogen. The water balance uses a simple moisture extraction routine based on increasing supply with increasing depth of extraction over time (Ritchie, 1972). Modifications to the OZCOT model to simulate rainfed single and double skip cotton production systems are described in Milroy et al. (2004). The OZCOT model has been independently validated against field measurements (Bange et al., 2005; Carberry and Bange, 1998; Conaty et al., 2018; Darbyshire et al., 2020; Milroy et al., 2004; Richards et al., 2008) for both rainfed and irrigated cotton crops (including commercial crops), and cotton simulations have been tested, and were found to be representative, at several sites,

including those chosen for investigation in this study. The planting date was set to 1st October with a row spacing of 1 m, with 14 plants/m, sown at a depth of 5 cm. Each crop started with an initial application of 300 kg/ha of N fertilizer, with an additional 50 kg/ha of N fertilizer applied 354 days after emergence. The soil water holding capacity was approximately 220 mm to a depth of 1.5 m. For irrigated scenarios, simulations started with a full soil profile and irrigation was automated to be non-limiting. Dryland scenarios were identical to irrigated scenarios, based on an assumption that there was water fully available at 30 cm and no water available at depth, and using rainfall based on the climate data each season. ET_0 (potential evapotranspiration (mm)) was assessed using primary pan data from 1970 onwards.

2.3. Analysis of trends

Linear regression models were used to analyse the climate variables as well as key growth outputs from OZCOT for significance (at $P < 0.05$) using R (R Core Team, 2021). Significance was reported for any trend over each 30 year climate normal and for the full 60 year period investigated.

3. Results

3.1. Changes in the frequency of extreme temperatures

There has been a decrease in the frequency of early season cold shocks at Bellata, Emerald, Griffith, Moree and Narrabri from 1961 to 2020 (Fig. 1, Supplementary Figure 1). This is combined with an increase in the seasonal minimum (night) temperatures across seven of the locations assessed (Supplementary Table). Despite no significant trend in the frequency of cold shocks at all locations from 1961 to 1990, there was an increase in night temperatures at Dalby, Emerald, Goondiwindi,

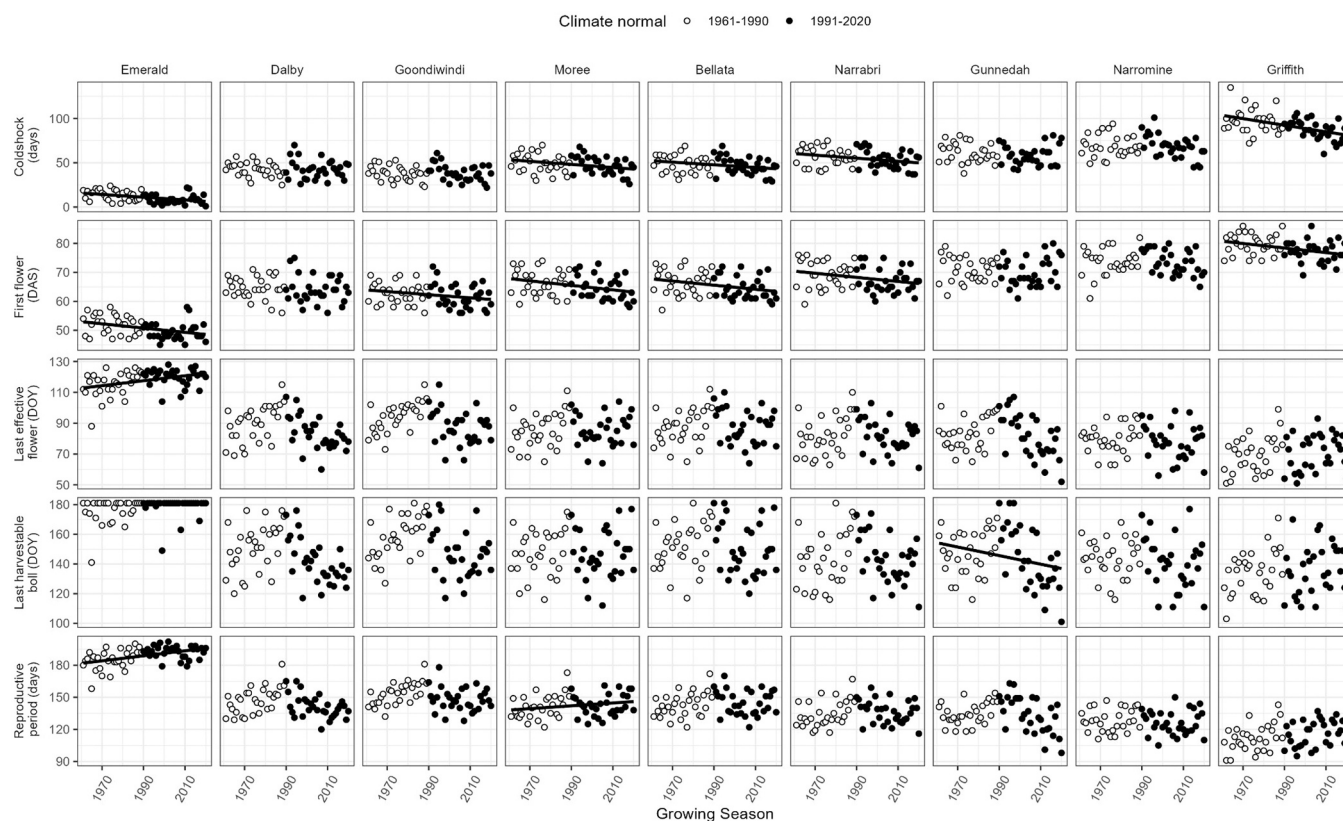


Fig. 1. Developmental trends for nine key Australian cotton locations from 1961 to 1990 (white symbols) and 1991–2020 (black symbols). Regression lines are shown where there is a significant trend at $P < 0.05$ across the 1961–2020 period. For rates of change, please refer to Supplementary Table.

and Moree. However, from 1991 to 2020, there were fewer instances of cold shock combined with warmer night temperatures at Bellata, Goondiwindi, Griffith, Moree, and Narromine.

At all nine locations, there was an increase in the average number of hot days above 40 °C from 1961 to 2020 (Fig. 2, Fig. 3). Eight locations (Bellata, Dalby, Goondiwindi, Griffith, Gunnedah, Moree, Narrabri, and Narromine) have had an increase in the number of days >40 °C from 1991 to 2020. There has also been a greater frequency of very hot days (>45 °C), especially in the more southern regions (i.e., Griffith and Narromine) from 1991 to 2020 compared with the period from 1961 to 1990 (Fig. 3).

3.2. Trends in seasonal temperature, ET_0 and crop development

Overall, the trends in DD_{12} and DD_{1532} were identical, with exception of an increase in DD_{32} at Moree during the 1961–1990 period compared with no change in DD_{12} (Fig. 2, Supplementary Figure 2). From 1961–2020, there has been an increase in the number of DD across most locations except for Dalby and Gunnedah. During the period 1961–1990, there was an increase in the number of DD throughout the northern cotton regions: Dalby, Emerald, Goondiwindi, and Moree. From 1991–2020, there was an increase in the number of DD at Bellata, Goondiwindi, Griffith, Moree, Narrabri, and Narromine. The number of DD vary across cotton regions with greater seasonal heat accumulation in northern regions than southern regions, for example 2322–2634 DD_{1532} in Emerald compared with 1528–1988 DD_{1532} in Griffith (Fig. 4). Our data shows not only an increase in the number of seasonal DD across many of Australia's key cotton regions, but also demonstrates how regions have shifted over time. As an example, the number of DD in Griffith prior to 2009 had been below 1900 DD_{32} , which had frequently occurred in Narrabri in the past. However, since 2009 there have been several years where the total number of DD_{1532} in Griffith have exceeded

1900, with a significant trend suggesting further increases in the number of DD with projected climate change.

Our data shows an increase in ET_0 across locations and time periods (Fig. 2), specifically with increased ET_0 at Griffith during the 1991–2020 period, and at Dalby, Griffith, and Narrabri over the 1961–2020 period (Supplementary Table). Whilst there has been an increase in ET_0 at these locations, it is also important to note that there have not been any significant changes in rainfall across any of the cotton regions or time periods.

Warmer seasonal temperatures have reduced the time to first flower across several locations from 1961 to 2020, and from 1961 to 1990, increasing the time to LEF, thereby lengthening the reproductive period of many cotton locations (Fig. 1). Warmer temperatures have not directly aligned with the trend for LHB to occur later in the season, due to increased variability during the period from 1991 to 2020 having resulted in either no evident changes in LHB, or a reduction in time to LHB in more recent years. Similarly, there have been variable climatic effects on reproductive period. Reproductive period increased at seven locations from 1961 to 1990, and increased at Emerald and Moree over the period 1961–2020. However, reductions in reproductive period were observed at Dalby and Gunnedah for the 1991–2020 time period.

3.3. Trends in OZCOT modelling

Using OZCOT, first flower was predicted to occur earlier in several locations for both irrigated and dryland systems (Table 1). Over the 1961–2020 and 1991–2020 time periods, first flower is predicted to occur earlier at Bellata, Griffith, Gunnedah, Moree, Narrabri, and Narromine in irrigated systems. During the 1961–2020 period, first flower was predicted to occur earlier in all nine locations in dryland systems, and at seven locations during the 1991–2020 period. Overall, there were few changes in lint yield. During the time periods 1961–2020, lint yield

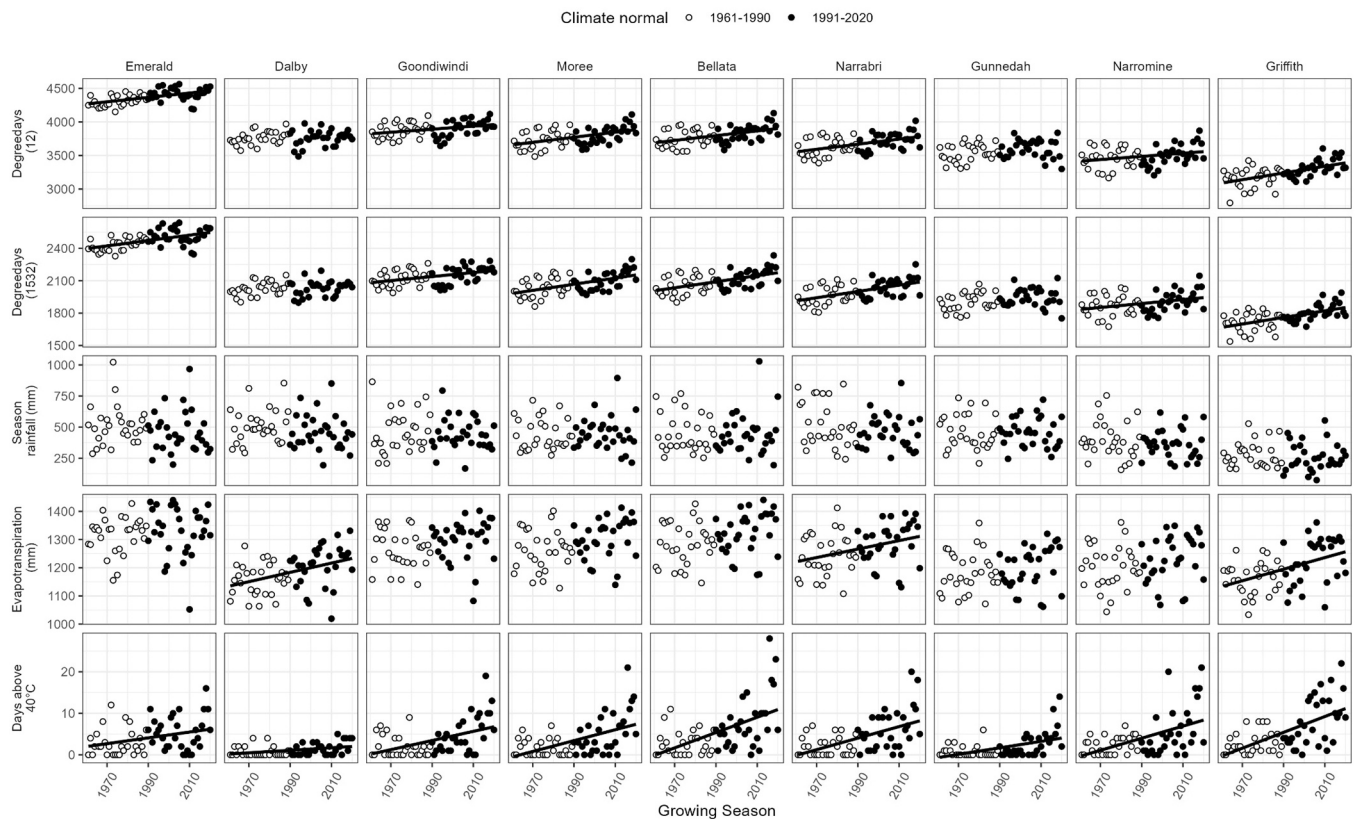


Fig. 2. Environmental trends for nine key Australian cotton locations from 1961 to 1990 (white symbols) and 1991–2020 (black symbols). Regression lines are shown where there is a significant trend at $P < 0.05$ across the 1961–2020 period. For rates of change, please refer to Supplementary Table.

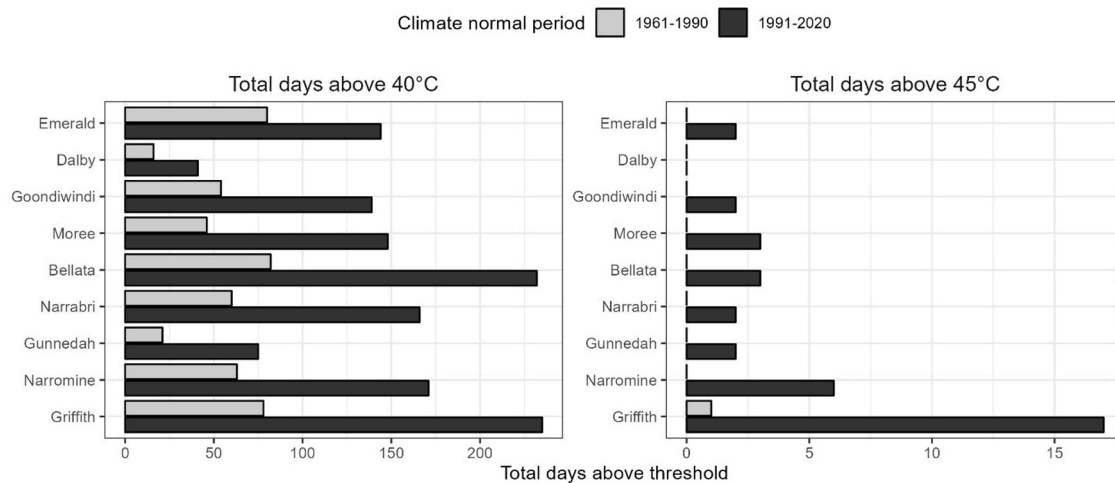


Fig. 3. The number of hot days (a) $>40^{\circ}\text{C}$ and (b) $>45^{\circ}\text{C}$ from 1st September to 30th April across various time periods. Values represent the total number of days during each climate normal period (1961–1990, grey and 1991–2020, black).

decreased in Narrabri for both irrigated and dryland systems. Additionally, lint yield decreased at Goondiwindi and Narrabri in irrigated systems during the 1991–2020 time period. Overall, there were few predicted changes in ET. During the 1961–2020 time period, in irrigated systems there were reductions in ET at Gunnedah and Narrabri, and in dryland systems there were reductions in ET at Dalby. During the 1991–2020 time period, reductions in ET were predicted in irrigated systems at Dalby and Goondiwindi. Additionally, there were few changes in WUE. During the 1961–1990 period, WUE was lower at Goondiwindi and Narrabri in dryland systems. During the 1991–2020 period, WUE was lower at Narrabri in irrigated systems.

4. Discussion

There have been climatic changes across each of the nine regions throughout the three time periods assessed from 1961 to 2020 (Fig. 1, Fig. 2), partially supporting our hypothesis. Overall, results show that there has been a decrease in the number of early season cold shocks, an increase in minimum night-time temperatures, an increase in the frequency of very high temperatures ($>40^{\circ}\text{C}$), an increase in season long heat accumulation (DD) and an increase in ET_0 at three locations. Overall, there has been a decrease in the time to first flower, but inconsistent trends in LEF, LHB and overall reproductive period between the 1961–1990 and 1991–2020 climate normal periods.

4.1. Frequency of extreme temperatures and minimum temperature changes

There has been a decrease in the frequency of early season cold shocks at Emerald, Griffith, Moree and Narrabri from 1961 to 2020 (Fig. 1). This is combined with an increase in the seasonal minimum (night) temperatures across six locations assessed (Supplementary Table). These findings reflect the State of the Climate (2022) report stating the frequencies of extremely cold days and nights have declined across Australia. Our study suggests that some key cotton regions are experiencing warmer temperatures, or less risk of very cold days that may slow the growth and development at the beginning of the season, thereby enabling the opportunity to extend the cotton season by planting earlier. Rising night temperatures also may have implications for increased night respiration rates reducing leaf ATP and carbohydrate content (Loka and Oosterhuis, 2010), reducing growth (Reddy et al., 1991a) and potentially lowering yield.

At all nine locations, there was an increase in the average number of hot days above 40°C from 1961–2020 (Fig. 2, Fig. 3). Six locations

(Dalby, Goondiwindi, Griffith, Gunnedah, Narrabri, and Narromine) have had an increase in the number of days $>40^{\circ}\text{C}$ from 1991 to 2020. There has also been a greater frequency of very hot days ($>45^{\circ}\text{C}$), especially in the more southern regions (i.e., Griffith and Narromine) from 1991 to 2020 compared with the period from 1961–1990 (Supplementary Table). Furthermore, in the coming decades, Australia is projected to experience continued warming, with more extremely hot days and fewer extremely cool days across many regions of the south and east (CSIRO and Bureau of Meteorology, 2022), encompassing Australian cotton regions. Increased frequencies of very hot temperatures during the cotton fruiting period may result in fruit loss (Reddy et al., 1991b), and consequently reduced yield especially if water stress is encountered. High air temperatures ($\geq 35^{\circ}\text{C}$) will also have a negative effect on physiology, productivity, and water use efficiency in cotton (Broughton et al., 2017b; Reddy et al., 1992).

4.2. Seasonal trends in temperature, ET_0 and crop development

From 1961–2020, there has been an increase in the number of DD across several locations (Fig. 2, Fig. 4). The implications of warmer temperatures are also reflected by an increase in ET_0 across locations and time periods (Fig. 2), highlighting the direct and indirect effects of warmer seasonal temperatures for both growth and development of the cotton and increased evaporation rates from the soil surface, potentially having led to a reduction in plant available soil water, despite no changes in seasonal rainfall. The temperature findings are consistent with the State of the Climate (2022) report stating that the long-term warming trend means that most years are now warmer than almost any observed during the 20th century, although the report also suggests that since the late 1990s across south-east Australia there has been a 10 % decline in April to October rainfall. These months largely fall outside the Australian cotton production season and thus are not captured in our study, however, are important considerations for water availability in the farming system and subsequent sustainability of cotton production in a changing climate (Bange et al., 2016). Frequent droughts and limited water resources for irrigation remains an important consideration, such that there may be breeding opportunities to exploit traits beneficial for water-limited (Broughton and Conaty, 2022; Conaty et al., 2022; Devi and Reddy, 2020; Sinclair et al., 2017) and warmer temperature, elevated CO_2 (Broughton et al., 2017a) environments, for cultivars that are increasingly tolerant to abiotic stresses.

Warmer seasonal temperatures have reduced the time to first flower (FF) across several locations from 1961 to 2020 as well as increasing the time to LEF at Emerald, thereby lengthening the reproductive period of

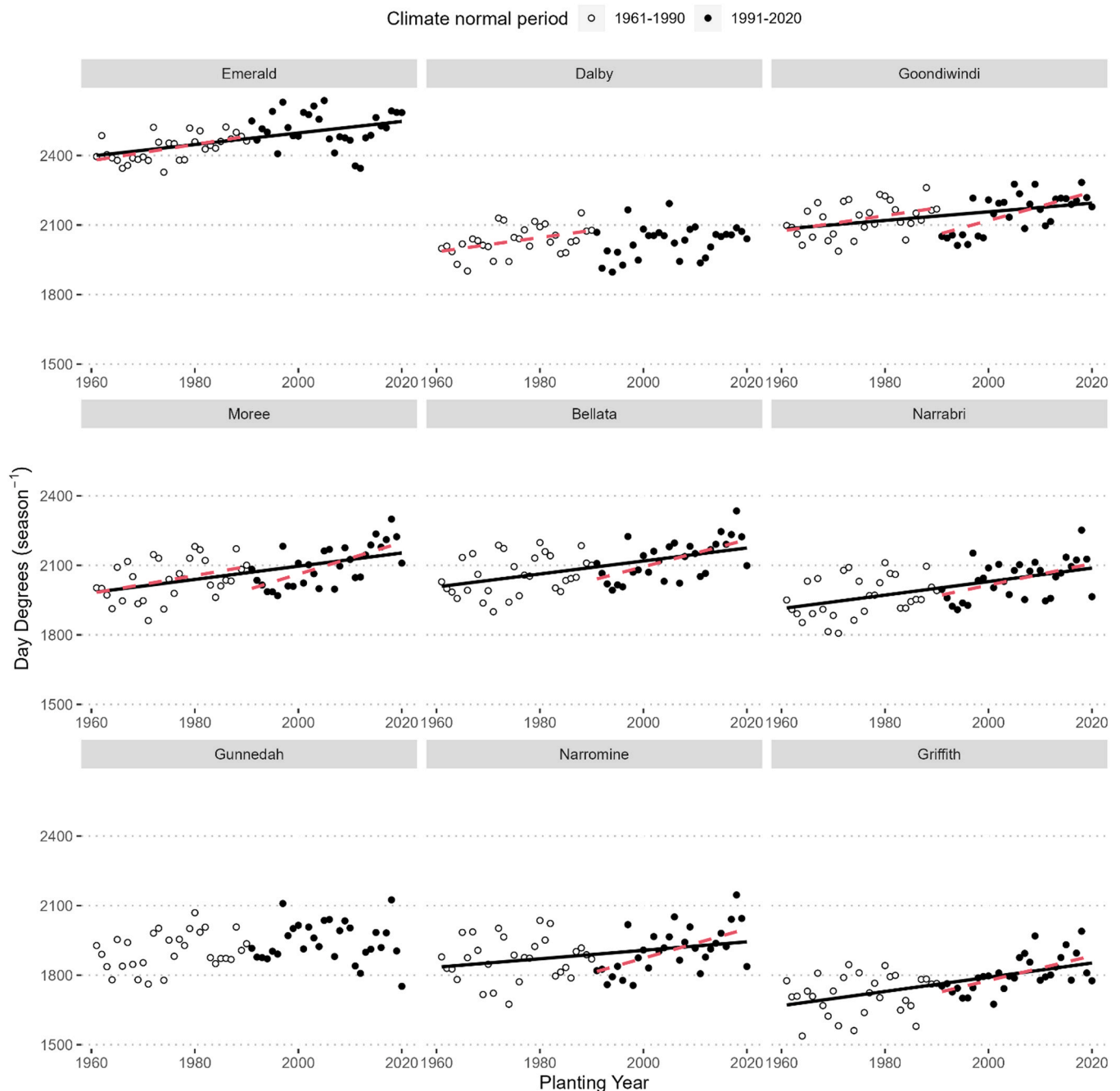


Fig. 4. The number of seasonal day degrees per year (DD_{1532}) across nine locations spanning the Australian cotton industry. Data is grouped by climate normal time periods (white symbols= 1961–1990; black symbols= 1991–2020). Regression lines are only shown where trends are significant at $P < 0.05$.

some cotton locations (Fig. 1, Table 1). Warmer temperatures have not directly aligned with the trend for LHB to occur later in the season, due to increased variability during the period from 1991–2020 having resulted in either no evident changes or reductions in LHB in more recent years. With warmer seasonal temperatures and the resulting effects on crop development, there has been an opportunity for increased yield across many Australian cotton regions whilst acknowledging that several studies have shown an increased yield attributed to improvements in genetics and management over this period (Liu et al., 2013). However, it is also possible that greater heat accumulation throughout the season and increased frequencies of high temperatures ($>35^{\circ}\text{C}$) may have resulted in increased vegetative biomass and greater requirements for the application of growth regulators such as mepiquat chloride. As such, our OZCOT modelling showed that overall, there were few

changes in lint yield for both irrigated and dryland crops and instances where lint yield predictions were reduced (Table 1). These instances of reduced lint yield were sometimes, but not always, associated with reductions in WUE. Therefore, recent improvements observed in cotton lint yields are more likely attributable to improvements in genetics and management rather than just climate change. Regardless, there will be a need to improve cotton farm resilience by maintaining and increasing cotton profitability through practices that increase both yield and fibre quality, while improving efficiency of resource use (especially energy, water and nitrogen) (Bange et al., 2016).

Our data shows not only an increase in the number of seasonal DD across many of Australia's key cotton regions, but also demonstrates how regions have shifted over time. As an example, the number of DD_{1532} in Griffith prior to 2009 had been below 1900 DD_{1532} , which had

Table 1

OZCOT predictions for irrigated and dryland systems across nine key Australian cotton locations and three time periods. ↓ indicates a significant increase, ↓ indicates a significant decrease, and (-) indicates there has been no significant trend at $P < 0.05$.

	Time Period & Location	FF	Irrigated			-	Dryland			
			Lint yield	ET	WUE		FF	Lint yield	ET	WUE
1961–2020	Emerald	-	-	-	-		↓	-	-	-
	Dalby	-	-	-	-		↓	-	↓	-
	Goondiwindi	-	-	-	-		↓	-	-	-
	Moree	↓	-	-	-		↓	-	-	-
	Bellata	↓	-	-	-		↓	-	-	-
	Narrabri	↓	↓	↓	-		↓	↓	-	-
	Gunnedah	↓	-	↓	-		↓	-	-	-
	Narromine	↓	-	-	-		↓	-	-	-
1961–1990	Griffith	↓	-	-	-		↓	-	-	-
	Emerald	-	-	-	-		-	-	-	-
	Dalby	-	-	-	-		-	-	-	-
	Goondiwindi	-	-	-	-		-	↓	-	↓
	Moree	-	-	-	-		-	-	-	-
	Bellata	-	-	-	-		-	-	-	-
	Narrabri	-	-	-	-		-	↓	-	↓
	Gunnedah	-	-	-	-		-	-	-	-
1991–2020	Narromine	-	-	-	-		-	-	-	-
	Griffith	-	-	-	-		-	-	-	-
	Emerald	-	-	-	-		-	-	-	-
	Dalby	-	-	↓	-		-	-	-	-
	Goondiwindi	-	↓	↓	-		↓	-	-	-
	Moree	↓	-	-	-		↓	-	-	-
	Bellata	↓	-	-	-		↓	-	-	-
	Narrabri	↓	↓	-	↓		↓	-	-	-
	Gunnedah	↓	-	-	-		↓	-	-	-
	Narromine	↓	-	-	-		↓	-	-	-
	Griffith	↓	-	-	-		↓	-	-	-

frequently occurred in Narrabri in the past (Fig. 4). However, since 2009 there have been several years where the total number of DD₁₅₃₂ in Griffith have exceeded 1900, with a significant trend suggesting further increases in the number of DD₁₅₃₂ with projected climate change. In this example, the knowledge of cotton systems in Narrabri may be relevant to the future management strategies of cotton grown in Griffith. This example highlights how improved knowledge of local climatic conditions combined with agronomic practices may be extended more broadly across different regions. Overall, there will be likely need to be a multi-faceted system-based approach combining many elements to harness climatic changes that are occurring and best enabling cotton industries to adapt (Bange et al., 2016), thereby ensuring sustainable cotton production into the future.

5. Conclusion

Climatic changes including decreased frequency of early season cold shocks, increased minimum temperatures and frequency of high temperatures and evapotranspiration, and an increase in season long heat accumulation have occurred across Australian cotton regions from 1961 to 2020, with implications for growth and development of cotton. Overall, warmer seasonal temperatures have reduced the time to first flower, increased the time to LEF, and resulted in some changes to LHB. This has resulted in a longer reproductive period and thus a greater opportunity for increased yield across Australian cotton regions. However, with these climatic advantages it is also important to acknowledge that there may have been concurrent disadvantages due to an increased frequency of extreme climatic conditions that likely resulted in fruit loss and increased ET_O, especially throughout the 1991–2020 period, where the reproductive period was reduced across some locations. Our results highlight that growers have already been contending with the effects of a changing climate and these adaptations and learnings are likely to be transferable across regions.

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CRediT authorship contribution statement

Michael Bange: Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Katrina Broughton:** Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Christopher Nunn:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Katrina Broughton reports financial support was provided by Australian Government Cotton Research and Development Corp.

Data Availability

The authors do not have permission to share data.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.fcr.2024.109491](https://doi.org/10.1016/j.fcr.2024.109491).

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